Flywheel Variable = X_w Frisbee/Disc Variable = X_d

Assuming **no slip condition**, tangential velocities are equal:

$v_w = v_d = v \quad \rightarrow \quad R_w\omega_w = R_d\omega_d \quad \rightarrow \quad v = R_w\omega_w \quad \rightarrow \quad \frac{\omega_d}{\omega_w} = \frac{R_w}{R_d}$

Thus a bigger flywheel means a lower required RPM to get the same exit velocity of the frisbee. Additionally, the larger the flywheel the greater the spin of the frisbee.

Breaking down the energy stored in the flywheel:

$E_w = \frac{1}{2} I_w \omega_w^2$

Flywheel inertia for a disc (replace with actual inertia value):

$I_w = \frac{1}{2} m_w R_w^2$

$E_w = \frac{1}{4} m_w R_w^2 \omega_w^2 \quad \rightarrow \quad E_w = \frac{1}{4} m_w v^2$

Energy of the frisbee:

$E_d = \frac{1}{2} m_d v^2 + \frac{1}{2} I_d \omega_d^2 \quad \rightarrow \quad I_d = \frac{1}{2} m_d R_d^2 \quad \rightarrow \quad E_d = \frac{3}{4} m_d v^2$

Required RPM:

$RPM = \frac{v}{2\pi R_w} * 60$

Assuming we want no more than a 20% Energy drop:

$E_w > 5E_d \quad \rightarrow \quad m_w > 15m_d$

Now looking at the motor torque requirements:

$\tau = I_w \alpha \quad \rightarrow \quad \alpha = \frac{\Delta\omega}{t_{spinup}} \quad \rightarrow \quad \tau = I_w * \frac{\Delta\omega}{t_{spinup}}$

Assuming a spinup time of 10 seconds:

$\tau = I_w * \frac{\Delta\omega}{10} \quad \rightarrow \quad \tau = \frac{1}{2} m_w R_w^2 * \frac{\Delta\omega}{10}$

Less sure about relevance of this part:

Torque exerted on motor:

$\tau \approx \frac{\Delta E}{\Delta\Theta}$

So for a quarter circle track:

$\tau \approx \frac{3m_d v^2/4}{\pi/2} = \frac{3}{2\pi} m_d v^2$

Rough MATLAB Code:

```
clc; clear; close all;
%% INPUT PARAMETERS (EDIT THESE)
v = 15;                 % Desired frisbee exit velocity (m/s)
Rw = 0.15;              % Flywheel radius (m)
Rf = 0.135;             % Frisbee radius (m) ~ standard ultimate disc
mw = 2.5;               % Flywheel mass (kg)
mf = 0.175;             % Frisbee mass (kg)
t_spinup = 10;           % Time to spin up flywheel (s)
%% 1) FLYWHEEL ANGULAR VELOCITY
omega_w = v / Rw;       % rad/s
RPM_w = omega_w * 60 / (2*pi);
%% 2) FRISBEE SPIN RATE
omega_f = v / Rf;       % rad/s
RPM_f = omega_f * 60 / (2*pi);
%% 3) FLYWHEEL INERTIA (solid disc)
Iw = 0.5 * mw * Rw^2;
%% 4) ENERGY PER SHOT (frisbee)
E_frisbee = (3/4) * mf * v^2;
%% 5) REQUIRED MOTOR TORQUE FOR SPIN-UP
alpha = omega_w / t_spinup;     % angular acceleration
tau_motor = Iw * alpha;         % Nm
%% 6) RPM DROP PER SHOT
% Fractional drop
frac_drop = (3/2) * (mf / mw);
% Absolute drop
delta_omega = frac_drop * omega_w;
delta_RPM = delta_omega * 60 / (2*pi);
%% DISPLAY RESULTS
fprintf('--- RESULTS ---\n');
fprintf('Flywheel RPM: %.1f RPM\n', RPM_w);
fprintf('Frisbee RPM: %.1f RPM\n\n', RPM_f);
fprintf('Flywheel inertia: %.4f kg*m^2\n', Iw);
fprintf('Energy per shot: %.2f J\n\n', E_frisbee);
fprintf('Motor torque (spin-up): %.2f Nm\n\n', tau_motor);
fprintf('RPM drop per shot: %.1f RPM\n', delta_RPM);
fprintf('Percent drop: %.1f %%\n', frac_drop * 100);
%% 7) TORQUE DURING LAUNCH (approx)
theta_contact = pi/2;   % quarter circle
tau_launch = E_frisbee / theta_contact;
fprintf('\nEstimated launch torque: %.2f Nm\n', tau_launch);
```

```
--- RESULTS ---
Flywheel RPM: 954.9 RPM
Frisbee RPM: 1061.0 RPM

Flywheel inertia: 0.0281 kg*m^2
Energy per shot: 29.53 J
```

Motor torque (spin-up): 0.28 Nm

RPM drop per shot: 100.3 RPM
Percent drop: 10.5 %

Estimated launch torque: 18.80 Nm